

GenCC Submission Portal API Guide

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GenCC Submission Portal API Guide

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Overview

The GenCC Submission Portal API enables programmatic submission and management of gene-disease curation records. This RESTful API allows authorized GenCC member organizations to:

- Submit new gene-disease curations
- Update existing submissions
- Query submission and job status

- Remove submissions

All submissions go through the same validation and release process as portal-submitted records.

Note: Spreadsheet file uploads are only available through the web portal interface. This API supports JSON-based submissions only.

Authentication

All API requests require authentication via an API key.

API Key Header

Include your API key in the request header:

GEN-API-KEY: your-api-token-here

Obtaining an API Key

API keys are automatically generated for each registered user. To view or regenerate your API key:

1. Log in to the GenCC Submission Portal
2. Navigate to your Profile settings
3. Your API key is displayed in the API Access section

Important: Keep your API key secure. Do not share it or commit it to version control.

Base URL

Production: `https://portal.thegence.org/api`

Submission API

Create Submissions

Submit one or more gene-disease curation records.

Endpoint: POST /api/submit

Headers:

Content-Type: application/json
GEN-API-KEY: your-api-token

Request Body:

```
{  
  "action": "create",
```

```

"submitter": {
  "id": "GENCC:000102"
},
"data": [
  {
    "action": "new",
    "gene": {
      "id": "HGNC:1234"
    },
    "disease": {
      "id": "MONDO:0012345"
    },
    "moi": {
      "id": "HP:0000006"
    },
    "classification": {
      "id": "GENCC:100001"
    },
    "report": {
      "display_date": "2026-02-15",
      "ext_url": "https://example.org/report/123"
    },
    "criteria": {
      "name": "ClinGen Gene-Disease Validity SOP",
      "url": "https://clinicalgenome.org/docs/summary-of-updates-to-the-clingen-gene-disease-clinical-validity-framework/"
    },
    "evidence": [
      { "pmid": "12345678" },
      { "pmid": "23456789" }
    ],
    "notes": {
      "display": "Public notes visible on GenCC website",
      "private": "Internal notes not published"
    },
    "contributors": {
      "primary": {
        "id": "contributor-id",
        "name": "Primary Contributor Name"
      }
    },
    "local_key": "your-internal-id-123"
  }
]
}

```

Success Response (200):

```
{
  "date": "2026-02-15T10:30:00Z",
  "message": "Job accepted",
  "jobs": [
    {
      "id": "J-100045",
      "message": "Job accepted",
      "status_code": 200
    }
  ]
}
```

Validate Submissions (Dry Run)

Validate submission data without creating records. Use this to check for errors before actual submission.

Endpoint: POST /api/submit/check

Headers:

Content-Type: application/json
GEN-API-KEY: your-api-token

Request Body: Same as Create Submissions

Success Response (200):

```
{
  "success": "true",
  "status_code": 200,
  "message": "OK"
}
```

Query Job Details

Retrieve full details of a job including all submissions.

Endpoint: GET /api/submit/job/{job_id}

Alternative: GET /api/query/job/{job_id}

Headers:

GEN-API-KEY: your-api-token

Response (200):

```
{
  "action": "status",
```

```

"type": "Job",
"jid": "J-100045",
"submitted": "2026-02-15T10:30:00Z",
"submitter": "CUID:300001",
"last_update": "2026-02-15T10:35:00Z",
"message": {
  "errorCode": null,
  "severity": "info",
  "text": "Your GenCC submission job processing status is:
    Staged"
},
"data": [
  {
    "type": "Submission",
    "submission_id": "SGC-100123",
    "submission_label": "BRCA1 / Breast cancer",
    "local_key": "your-internal-id-123",
    "submitted": "2026-02-15T10:30:00Z",
    "last_update": "2026-02-15T10:35:00Z",
    "status": "Staged",
    "gene": {
      "id": "HGNC:1100",
      "symbol": "BRCA1"
    },
    "disease": {
      "id": "MONDO:0007254",
      "name": "breast cancer"
    },
    "moi": {
      "id": "HP:00000006",
      "name": "Autosomal dominant"
    },
    "classification": {
      "id": "GENCC:100001",
      "name": "Definitive"
    }
  }
]
}

```

Query Submission Details

Retrieve details of a specific submission.

Endpoint: GET /api/submit/{submission_id}

Alternative: GET /api/query/{submission_id}

Headers:

GEN-API-KEY: your-api-token

Response: Similar to job query but for a single submission.

Check Job Status

Get the current processing status of a job.

Endpoint: GET /api/status/job/{job_id}

Alternative: GET /api/submit/job/{job_id}/status

Headers:

GEN-API-KEY: your-api-token

Response (200):

```
{
  "date": "2026-02-15T10:40:00Z",
  "message": "Status Response",
  "jobs": [
    {
      "id": "J-100045",
      "message": "Staged",
      "status_code": 200
    }
  ]
}
```

Check Submission Status

Get the current status of a specific submission.

Endpoint: GET /api/status/{submission_id}

Alternative: GET /api/submit/{submission_id}/status

Headers:

GEN-API-KEY: your-api-token

Response (200):

```
{
  "date": "2026-02-15T10:40:00Z",
  "message": "Status Response",
  "jobs": [
    {
```

```
{
  "id": "J-100045",
  "message": "Staged",
  "status_code": 200,
  "data": [
    {
      "sid": "SGC-100123",
      "message": "Published",
      "status_code": 200
    }
  ]
}
```

Remove Job

Remove a job and all its associated submissions.

Endpoint: GET /api/remove/job/{job_id}

Alternative: GET /api/submit/job/{job_id}/remove

Headers:

GEN-API-KEY: your-api-token

Response (200):

```
{
  "success": "true",
  "status_code": 200,
  "message": "Job Removed"
}
```

Remove Submission

Remove a specific submission.

Endpoint: GET /api/remove/{submission_id}

Alternative: GET /api/submit/{submission_id}/remove

Headers:

GEN-API-KEY: your-api-token

Response (200):

```
{
  "success": "true",
```

```
"status_code": 200,  
"message": "Submission Removed"  
}
```

Data Schema

Submission Object

Field	Type	Required	Description
action	string	Yes	"new" for new submissions, "update" for updates
gene.id	string	Yes	HGNC gene identifier (e.g., "HGNC:1234" or "1234")
disease.id	string	Yes	Disease identifier (MONDO, OMIM, or ORPHA)
moi.id	string	Yes	Mode of inheritance HPO term (e.g., "HP:0000006")
classification.id	string	Yes	GenCC classification ID (e.g., "GENCC:100001")
report.display_date	string	Yes	Evaluation date (YYYY-MM-DD format)
criteria.url	string	Yes	URL to assertion criteria documentation
criteria.name	string	No	Name of assertion criteria
report.ext_url	string	No	URL to external report
evidence	array	No	Array of PubMed references
evidence[].pmid	string	No	PubMed ID
notes.display	string	No	Public notes (visible on GenCC website)
notes.private	string	No	Private notes (not published)
contributors.primary.name	string	No	Primary contributor name
contributors.primary.id	string	No	Primary contributor identifier
local_key	string	No	Your internal tracking identifier
mechanism.id	string	No	Mechanism of disease (e.g., "GENCC:200001")
mechanism.comment	string	No	Comment on mechanism

Required Fields

For new submissions, the following fields are **required**: - `gene.id` - Valid HGNC gene identifier - `disease.id` - Valid disease identifier (MONDO preferred, OMIM/ORPHA accepted) - `moi.id` - Valid HPO mode of inheritance term - `classification.id` - Valid GenCC classification - `report.display_date` - Evaluation date - `criteria.url` - Assertion criteria URL

Reference IDs

Submitter IDs

Use your organization's GenCC submitter ID. Contact GenCC to obtain your submitter ID if you don't have one.

Classification IDs

ID	Classification
GENCC:100001	Definitive
GENCC:100002	Strong
GENCC:100003	Moderate
GENCC:100004	Limited
GENCC:100005	Disputed Evidence
GENCC:100006	Refuted Evidence
GENCC:100007	Animal Model Only
GENCC:100008	No Known Disease Relationship
GENCC:100009	Supportive

Mode of Inheritance IDs (Common)

ID	Mode of Inheritance
HP:0000006	Autosomal dominant
HP:0000007	Autosomal recessive
HP:0001417	X-linked
HP:0001419	X-linked recessive
HP:0001423	X-linked dominant
HP:0001427	Mitochondrial
HP:0032113	Semidominant
HP:0000005	Unknown

See the GenCC Submission Spreadsheet for the complete list of accepted values.

Error Codes

Code	HTTP Status	Description
9000	401	Missing API key

Code HTTP Status Description

9001	401	Unauthorized - invalid API key
9002	401	Unauthorized - inactive user
9003	401	Submitter mismatch - user not authorized for submitter
9005	401	Unauthorized - user not found
9006	401	Job not found or unauthorized
9007	401	Submission not found or unauthorized
1001	501	Invalid JSON format
1002	501	JSON schema error
1003	501	Unknown action
1004	501	Submission validation error

Examples

cURL: Create a New Submission

```
curl -X POST https://portal.thegenc.org/api/submit \
  -H "Content-Type: application/json" \
  -H "GEN-API-KEY: your-api-token-here" \
  -d '{
    "action": "create",
    "submitter": { "id": "GENCC:000102" },
    "data": [{
      "action": "new",
      "gene": { "id": "HGNC:1100" },
      "disease": { "id": "MONDO:0007254" },
      "moi": { "id": "HP:0000006" },
      "classification": { "id": "GENCC:100001" },
      "report": {
        "display_date": "2026-02-15",
        "ext_url": "https://example.org/report"
      },
      "criteria": {
        "name": "ClinGen SOP",
        "url": "https://clinicalgenome.org/sop"
      },
      "evidence": [{ "pmid": "12345678" }],
      "local_key": "internal-123"
    }]
  }'
```

cURL: Check Job Status

```
curl -X GET https://portal.thegenc.org/api/status/job/J-100045 \
  -H "GEN-API-KEY: your-api-token-here"
```

Python Example

```
import requests

API_BASE = "https://portal.thegenc.org/api"
API_KEY = "your-api-token-here"

headers = {
    "Content-Type": "application/json",
    "GEN-API-KEY": API_KEY
}

# Create submission
payload = {
    "action": "create",
    "submitter": {"id": "GENCC:000102"},
    "data": [{
        "action": "new",
        "gene": {"id": "HGNC:1100"},
        "disease": {"id": "MONDO:0007254"},
        "moi": {"id": "HP:0000006"},
        "classification": {"id": "GENCC:100001"},
        "report": {
            "display_date": "2026-02-15",
            "ext_url": "https://example.org/report"
        },
        "criteria": {
            "url": "https://clinicalgenome.org/sop"
        }
    }]
}

response = requests.post(f"{API_BASE}/submit", json=payload,
                        headers=headers)
print(response.json())

# Check status
job_id = "J-100045"
status = requests.get(f"{API_BASE}/status/job/{job_id}",
                    headers=headers)
print(status.json())
```

Support

Technical Support

For API issues, authentication problems, or error troubleshooting: - **Email:** gencc-tech@broadinstitute.org

General Inquiries

For scientific questions, data inquiries, or GenCC participation: - **Email:** gencc@thegencc.org

Additional Resources

- [GenCC Website](#)
- [GenCC Submission Portal](#)
- [User Guide](#)